

[서식1] 학생 작성용

캡스톤디자인 과제계획서

과 제 명	where-is-it			
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(Oiiil7)팀은 2026학년도 2학기 캡스톤디자인 교과목 과제를 성실하게 수행하고자 위와 같이 과제계획서를 제출하며, 중도 포기하거나 과학기술정보통신부의 지적 및 환수 시, 제반 규정에 따 라 과제 수행을 위해 지원받은 지원금을 전액 반납할 것을 서약합니다. 2026년 9 월 21 일 대표학생 <u>Shwe Yin Min Oo</u> (인) 지도교수 <u>윤대균</u> (인) 아주대학교 SW중심대학사업단장 귀하				

1. 과제의 개요

where-is-it : An AI driven Memory-aid Camera-based System for Facilitating Users in Locating Misplaced Objects

Background

People frequently misplace small everyday belongings—wallets, keys, phones, and glasses—at home or in the office. This happens for many reasons: they may forget where an item was placed, or it may have been moved by a pet or another person. Although tracking devices exist, most people do not bother to attach them to every item. Meanwhile, many homes and offices already have CCTV or security cameras that are largely unused beyond surveillance.

Purpose

This project aims to transform existing cameras into a passive memory assistant and object locator. Rather than simply answering, “Where is my wallet now?”, the system addresses the questions people actually ask when something goes missing where it was last seen, when it disappeared, what happened nearby, and where to look next—all through natural-language conversation.

Development Content

- Object Detection & Logging: Periodically captures camera frames, detects visible objects, and records their location, timestamp, and confidence.
- Location History Database: Stores object detections over time, allowing recent movement to be reconstructed rather than showing only the current location.
- Agent Reasoning: Uses a three-stage search process: current -> historical -> contextual.
- Natural-Language Interface: Allows users to ask questions naturally (e.g., “Where’s my wallet?”) and receive answers that clearly distinguish confirmed facts from inferred possibilities.

Necessity

Physical tracking devices are practical for a limited number of important items but are inconvenient for the many small objects people misplace every day. This project eliminates the need for per-item setup by using cameras that people already own. It can be particularly useful for people with memory difficulties while also helping anyone who occasionally struggles to find misplaced belongings.

2. 과제의 목표

Purpose

The primary goal of the system is to help users locate misplaced belongings efficiently. It achieves this by detecting the current locations of objects from camera feeds, searching historical location records, and analyzing temporal patterns and contextual clues to determine where an object may be located. Based on this information, the system provides users with practical and actionable suggestions to guide their search.

In addition, the system aims to reduce the cognitive load on users, particularly those who may experience memory difficulties. Rather than providing uncertain information as fact, it presents confidence-based answers to clearly communicate the reliability of its predictions. The system is also designed to learn from accumulated data and user interactions to improve its accuracy over time while maintaining strong privacy and security protections for users and their data.

Final Deliverable

Where-Is-It AI MVP Prototype

- Object Detection & History Logging Module — takes camera input (or pre-recorded images for MVP testing), detects objects, and automatically records location, time, and confidence.
- AI Agent (Natural-Language Q&A System) — searches detection history and generates answers in three stages: current location -> last confirmed location -> contextual analysis.
- Simple Chat Interface — lets users ask natural-language questions (e.g., “Where is my wallet?”) and receive speech-based answers
- Architecture and Ontology Documentation — documents describing the system's structure and mini ontology
- Demo video or live demo covering three real scenarios: current detection, historical lookup, and context-based inference

Evaluation Criteria

The Where Is It? system will be evaluated based on object detection accuracy, location tracking accuracy, response time, usability, and system reliability.

Quantitative Goals

- Object detection accuracy: ~90%
- Location tracking accuracy: ~85%
- Average response time: <3 seconds
- False-certainty rate: <2%
- Historical data: query able for the most recent 7 days

Qualitative Goals

The system should provide clear and useful information about misplaced items, offer an intuitive user experience, and provide reasonable suggestions when an object's exact location cannot be determined. The system will also prioritize the protection of camera data and location history and maintain an ethical and careful response style that does not assign blame to specific people or pets.

Industry Collaboration Strategy and Deliverable Utilization Strategy

During the project, we plan to obtain feedback from individuals and companies working in AI and smart home technology. Their feedback will be used to improve system functionality, usability, and applicability to real-world users.

After completion, the where-is-it AI system can be further developed as a smart home service for

locating lost or misplaced items. It could be offered as an add-on service within existing smart home ecosystems, such as home camera applications or smart home hubs, reducing the barrier to adoption.

The project can also serve as a foundation for future research and development, including improved object tracking, contextual reasoning, personalization, and integration with additional smart home devices.

3. 결과물에 대한 기대효과 및 활용방안

The where-is-it system is expected to help users find lost or misplaced items more easily and reduce the time spent searching for them. Through this project, we can also gain practical experience in AI, object detection, location tracking, and application development.

After completion, the system can be further improved and used in areas such as smart homes, elderly care, and personal assistance services. With further development, it may also have potential for commercialization as a smart home application or AI-based item tracking service.

4. 수행방법

The project will be developed step by step, starting with the basic system design and development environment setup. Cameras will be used to capture images or video, and an object detection model will identify and track items. The detected object information and location history will be stored in a database. The system will then connect the backend, AI functions, and user interface so that users can search for their belongings easily. After development, we will test the system in different situations, check its accuracy and response time, and address any problems found during testing.